

WEEK 6 – TUTORIAL

Objectives:

The objective of this tutorial is to enable students to apply **key quality management tools and techniques in practical IT project scenarios**, including analysing defects using cause-and-effect diagrams and checksheets, detecting patterns through quality tools, evaluating the Cost of Conformance and Cost of Non-Conformance, and understanding trade-offs between cost, time, and quality to recommend actions that optimize overall project quality and minimize total Cost of Quality.

Tutorial Activities:

- Tools and Techniques
- Cost Analysis and Trade Offs

Learning outcomes

By the end of this tutorial, students will be able to:

- **LO1.** Understand each of the core project management knowledge areas
- **LO2.** Apply project management knowledge, tools and techniques
- **LO4.** Develop basic proficiency in project management tools and practices

Part A: Tools and Techniques

Scenario 1: A cloud-based HRM system is experiencing frequent payroll calculation errors.

- Draw a cause-and-effect diagram for the payroll defect.
- Suggest three preventive measures using the diagram as a reference.
- Prepare a small checksheet to track similar defects in future releases.

Scenario 2: A module shows intermittent system crashes during report generation.

- Choose two quality management tools or techniques to identify trends or patterns. Explain why you chose them.

Part B: Cost Analysis and Trade Offs

Scenario: During testing, the QA team discovers defects costing \$8,000 to fix internally. Skipping some preventive measures would save \$2,000 now but could result in \$20,000 in external failure costs post-release.

- Identify the Cost of Conformance and Cost of Non-Conformance in this scenario.
- Discuss the trade-off between cost, time, and quality.
- Suggest one action the project manager should take to minimize total CoQ.